

Genetics

(Scheduled Event - 30 Minutes)



Description: Participants will be assessed on their understanding of genetics, heredity, inheritance patterns, and the relationship between genes and traits.

Participants per assigned Team Number: 2

- If your school has 1 team you will send 2 students
- If your school has 2 teams you will send 2 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 2 students per team number; team numbers may not intermix

Teams must bring: Pen/Pencil

Teams may not bring:

- Notes or guides of any kind.

Optional - teams may bring: Four-function calculator

Safety Requirements:

- None

The Competition: This event may be administered as a written test or as a series of stations which can include diagrams, models, pedigrees, Punnett squares, data analysis, and problems for students to solve. Content topics will include but are not limited to:

1. Fundamentals of Genetics

- Define and use the terms DNA, chromosome, gene, allele, genotype, phenotype, character, trait, homozygous, heterozygous, dominant, and recessive.
- Describe the relationship between DNA, genes, chromosomes, and inherited traits.
- Explain how genes are inherited from parents and how genotype influences phenotype.

2. DNA and Heredity

- Describe the basic structure and composition of DNA, including nucleotides and complementary base pairing.
- Explain how DNA stores genetic information.

3. Mendelian Genetics

- Understand Mendel's contributions to heredity.
- Explain dominant and recessive inheritance patterns.
- Construct and analyze Punnett squares for monohybrid crosses.
- Determine genotype ratios, phenotype ratios, and probabilities of inheritance.

4. Human Traits and Patterns of Inheritance

- Identify common inherited human traits, including earlobe attachment, tongue rolling, cleft chin, dimples, ACHOO syndrome, and colorblindness.
- Describe inheritance patterns, including autosomal dominant, autosomal recessive, and X-linked recessive inheritance.
- Understand that some traits are simplified classroom examples of Mendelian inheritance, while many human traits are influenced by multiple genes.
- Recognize ACHOO syndrome as an autosomal dominant trait and colorblindness as an X-linked recessive trait.

- Analyze pedigrees and inheritance scenarios to determine possible patterns of trait transmission.

5. **Beyond Simple Dominance**

- Recognize examples of codominance, incomplete dominance, multiple alleles, and sex-linked traits.
- Compare these inheritance patterns to simple dominant-recessive inheritance.

6. **Inherited Traits, Learned Behaviors, and Adaptations**

- Distinguish between inherited traits, instinctive behaviors, and learned behaviors.
- Identify examples of inherited characteristics and learned behaviors in plants and animals.
- Explain how inherited adaptations help organisms survive in specific environments.

7. **Data Analysis and Applications**

- Interpret diagrams, pedigrees, charts, tables, and models related to genetics and heredity.
- Analyze inheritance data and determine likely patterns of inheritance.
- Apply genetic concepts to solve problems and answer scenario-based questions.

Examples:

- Identify and define genetics vocabulary (DNA, gene, chromosome, allele, genotype, phenotype, dominant, recessive, homozygous, heterozygous).
- Interpret diagrams showing the relationship between **DNA → genes → chromosomes → inherited traits**.
- Analyze DNA models to identify nucleotides and complementary base pairs.
- Use DNA sequences or models to determine complementary DNA strands.
- Interpret Punnett squares to predict possible offspring genotypes and phenotypes.
- Calculate genotype ratios, phenotype ratios, and inheritance probabilities from monohybrid crosses.
- Determine whether individuals are homozygous or heterozygous based on genetic information.
- Analyze pedigrees to identify whether a trait is likely **dominant, recessive, autosomal, or X-linked**.
- Determine possible genotypes of family members from pedigree information.
- Predict inheritance patterns for common classroom traits (e.g., earlobe attachment, tongue rolling, dimples, cleft chin).
- Analyze scenarios involving **ACHOO syndrome** as an autosomal dominant trait.
- Analyze scenarios involving **colorblindness** as an X-linked recessive trait.
- Compare inheritance patterns including:
 - simple dominant/recessive inheritance
 - incomplete dominance
 - codominance
 - multiple alleles
 - sex-linked traits
- Interpret tables comparing parental traits and offspring outcomes.
- Analyze case studies to determine whether a characteristic is inherited, learned, or an adaptation.
- Distinguish between:
 - inherited traits (eye color, blood type)
 - instinctive behaviors (bird migration, spider webs)
 - learned behaviors (language, trained behaviors)
- Apply genetics concepts to real-world scenarios involving family inheritance, animal breeding, and plant traits.
- Analyze how inherited adaptations help organisms survive in different environments.

Scoring:

- Points will be awarded for each correct response.

Tiebreakers:

- Ties will be broken by the accuracy or quality of answers to select questions chosen by the event leader prior to the competition.

Resources:

- [Biology for Kids:Genetics](#)
- Science Teachers & Librarians
- Get permission from your parents/coach to use AI to create a study guide using the rules provided here

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles